

Tian Qin

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Research Interests

Model predictive control (MPC), adaptive/fuzzy PID, autonomous mining trucks and safety-critical autonomous systems, vehicle dynamics modeling and parameter identification, data-driven/reinforcement-learning-based control.

Education

China University of Mining and Technology, Beijing (211)

Sep. 2023 - Sep. 2025

M.S.: Control Science and Engineering | GPA: 3.59/4.0

Supervisor: Prof. Kehu Yang, Intelligent Energy Systems and Autonomous Mining Laboratory

Core coursework: Linear System Theory (96), Adaptive Control (94), Advanced Signal Processing (93)

Sichuan University (985)

Sep. 2015 - Jun. 2019

B.Eng.: Electronic Information Engineering

Core coursework: C Programming (91), Computer Communication and Networks (86), Communication Principles Laboratory (88), Analog Electronics, Signals and Systems

Publications

Journal

- **Tian Qin**, Desheng Zhu, Chunhui Wang, Peize Yang, Kehu Yang. Dual-Loop Fuzzy-PID Acceleration Tracking Controller for Autonomous Mining Trucks under Variable Payload Conditions[J]. Coal Engineering, 2025, 57(6): 172-179. DOI:10.11799/ce202506022.
Designed for longitudinal dynamics changes between empty and full-load autonomous mining trucks, this work develops a dual-loop fuzzy-PID acceleration tracking controller: an outer speed-tracking loop, an inner acceleration-control loop, and online PID adjustment according to payload state. Simulations show stable speed and acceleration tracking under variable-payload conditions, with a maximum speed error of 0.49 km/h and a maximum acceleration error of 0.103 m/s².

Conference

- **Tian Qin**, Lei Qiu, Jingchuan Chen, Hongyi Fu, Desheng Zhu, Kehu Yang. Double-Layer Following Controller for Autonomous Vehicles[C]//2024 36th Chinese Control and Decision Conference (CCDC). IEEE, 2024: 908-913. DOI:10.1109/CCDC62350.2024.10587700.
For high-speed overtaking, sharp turns, and dynamic obstacle avoidance in autonomous following, this work builds a two-layer following-control framework combining DWA local planning with fuzzy-adaptive PID. The upper layer plans a safe local path from lane markings and target-vehicle information, while the lower layer adapts PID parameters online to maintain safe distance and improve lateral/longitudinal smoothness.
- Wang C, Zhu D, **Tian Qin**, Yang K. Lateral Trajectory Tracking of Autonomous Mining Trucks Using MPC with Adaptive Load Compensation[C]//2025 44th Chinese Control Conference (CCC). IEEE, 2025. DOI:10.23919/CCC64809.2025.11178743.
To address reduced lateral tracking accuracy on complex open-pit mine roads under payload variation, this work proposes an MPC strategy with adaptive load compensation. The controller updates vehicle parameters in real time for empty/full-load states and optimizes control outputs to reduce the effect of payload fluctuation on prediction and trajectory tracking.
- Yang P, Zhu D, **Tian Qin**, Yang K. Adaptive Heading Tracking Algorithm Based on Vehicle Dynamics Model[C]//2025 44th Chinese Control Conference (CCC). IEEE, 2025. DOI:10.23919/CCC64809.2025.11179654.
To overcome pure pursuit's limited use of target-path heading information, this work designs an improved pure-pursuit algorithm with vehicle-dynamics feedback and heading correction. By considering vehicle states, environmental characteristics, and target-path direction, the method reduces average lateral error from 0.462 m to 0.237 m and improves tracking accuracy and adaptability on complex mine roads.

Research Experience

Key Technologies and Application Demonstration of Autonomous Transportation and Loading/Unloading System for Large Open-Pit Mine Robots

National Key Research and Development Program of China

Nov. 2022 - Oct. 2025

- Developed adaptive/fuzzy PID longitudinal-control algorithms for low-speed autonomous mining trucks, focusing on speed and acceleration tracking under variable-payload operating conditions.
- Extended this control foundation to load-adaptive MPC for lateral trajectory tracking by incorporating vehicle dynamics and payload-dependent parameters, improving robustness in open-pit mine scenarios.
- Formulated follow-up research problems around model calibration, parameter identification, and learning/data-driven control for safety-critical autonomous systems.

Quantitative Stock Prediction and Trading Strategy

Independent Project

- Processed historical market data for a selected stock, engineered features, and built prediction-modeling pipelines using MLP, random forest, and other machine-learning models.
- Combined model outputs through a weighted ensemble and converted prediction signals into a rule-based buy/sell strategy.
- Backtested the strategy over the selected test period, achieving a 13% backtested return and 55% directional prediction accuracy.

Work Experience

Mindway Science and Technology Co., Ltd. - Beijing

Oct. 2025 - Present

Control Algorithm Engineer

- Developed and deployed lateral/longitudinal control code for autonomous mining trucks: LQR for lateral control and dual-loop PID for longitudinal control, covering path tracking, speed tracking, and braking control; completed tuning and validation on a real-vehicle platform.
- Independently implemented AEB anti-collision logic based on time-domain trajectory prediction and OBB detection, judging collisions between the ego vehicle's future trajectory and obstacle bounding boxes for safety-critical mine-road scenarios.
- Performed system identification using real-vehicle data, identifying parameters of a two-degree-of-freedom vehicle dynamics model to provide accurate model parameters for LQR tuning and vehicle validation.

China Aerospace Science and Industry Corporation (CASIC) - Wuhan

Jul. 2019 - Aug. 2023

Hardware Engineer

- Participated in industrial-vehicle control-system calibration and on-site debugging, including engine-throttle calibration.
- Coordinated joint debugging among vehicle controllers, sensors, and actuators; adjusted control parameters and interface configurations to resolve response lag, communication anomalies, and on-site adaptation issues during calibration.
- Supported on-site testing, data logging, and issue reproduction, feeding calibration results back into control-strategy and hardware-configuration iterations to support stable operation after vehicle retrofit.

Selected Competitions

China Intelligent Connected Vehicle Challenge

Oct. 2023 - Nov. 2023

Contribution Award (Top 5%)

- For adaptive cruise and following tasks, used DWA for local path planning and fuzzy PID for speed/acceleration tracking; evaluated comfort using lateral/longitudinal acceleration RMS, achieving values below 0.10 m/s² and 0.06 m/s², respectively.

Autopilot Challenge of Vehicle-Road-Cloud Integration

Mar. 2024 - Jun. 2024

Innovation Award (Top 5%, ranked 7th among 118 teams)

- Integrated and tuned an LQR lateral controller on a real autonomous-driving platform, computing control inputs from path error and vehicle states to support low-speed parking and obstacle-avoidance validation.

National Post-Graduate Mathematical Contest in Modeling

Oct. 2024 - Nov. 2024

Participation Award (Top 20%)

- For magnetic-component loss modeling, fitted Steinmetz-equation parameters with nonlinear optimization, analyzed the effect of operating-condition variables on model accuracy, and improved model accuracy by 5%.

Technical Skills and Languages

Programming: C++, Python, MATLAB

Autonomous driving and simulation: Apollo, ROS, Linux, VTD, TruckSim, CarSim

Control and modeling: MPC, LQR, fuzzy PID, vehicle dynamics, model calibration, parameter identification

Languages: Chinese, English (IELTS 6.5; Reading 8.0)